Literature Status and Explicit Counterexample: Projective Tensors Do Not Preserve Property-U

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Status. literature_already_answered. The later paper gives a negative answer; below is a direct two-dimensional witness.

Question. Daptari–Paul–Rao, arXiv:2009.09581, Theorem 3.7 proves under suitable hypotheses that property-U and property-SU pass back and forth between $Y \subset Z$ and $X \widehat{\otimes}_\varepsilon Y \subset X \widehat{\otimes}_\varepsilon Z$. Immediately afterward they ask whether the analogous conclusion holds for $\widehat{\otimes}_\pi$.

Later answer. Daptari–Paul, arXiv:2202.00947, states in its tensor-product subsection that property-U need not be stable under projective tensor products, citing Basu–Rao, Example 2.3. It then proves only the restricted positive result that, when Y has property-SU in Z, every *isometry* in $\mathcal{L}(X, Y^*)$ has a unique norm-preserving extension. Property-U would require this for every bounded operator.

Explicit finite-dimensional witness. Work over $\mathbb{R}$, as in the source, and take

$$X = \ell_\infty^2, \qquad Z = \ell_2^2, \qquad Y = \operatorname{span}\{e_1\}.$$

Here $X^* = \ell_1^2$, so X is an L_1-predual; Z^* has the RNP; and Y has property-SU in the Hilbert space Z. The orthogonal projection $Z \to Y$ is contractive, hence the canonical map $X \widehat{\otimes}_\pi Y \to X \widehat{\otimes}_\pi Z$ is an isometry.

Using $(X \widehat{\otimes}_\pi Z)^* = \mathcal{B}(X, Z)$, for $x = (a, b)$ and $z = (s, t)$ define

$$G_0(x, z) = \frac{s(a+b)}{2}, \qquad G_1(x, z) = \frac{s(a+b)}{2} + \frac{t(a-b)}{2}.$$

They have the same restriction $F(x, se_1) = s(a+b)/2$ to $X \times Y$. Since Y is one dimensional,

$$\|F\| = \|(1/2, 1/2)\|_{\ell_1^2} = 1, \qquad \|G_0\| = 1.$$

For the other extension, optimize first over $z \in B_{\ell_2^2}$:

$$\|G_1\| = \sup_{|a|,|b| \le 1} \sqrt{\left(\frac{a+b}{2}\right)^2 + \left(\frac{a-b}{2}\right)^2} = \sup_{|a|,|b| \le 1} \sqrt{\frac{a^2+b^2}{2}} = 1.$$

But $G_0 \neq G_1$ (evaluate at $x = (1, -1)$ and $z = e_2$). Thus F has two distinct norm-preserving extensions. Therefore

$$X \widehat{\otimes}_\pi Y \text{ does not have property-}U \text{ in } X \widehat{\otimes}_\pi Z.$$

Since property-SU implies property-U, this single example disproves both proposed projective-tensor equivalences, despite satisfying every hypothesis of the injective theorem.

References

[1] S. Daptari, T. Paul, and T. S. S. R. K. Rao, *Uniqueness of Hahn–Banach extension and related norm-1 projections in dual spaces*, arXiv:2009.09581.

[2] S. Daptari and T. Paul, *Uniqueness of Hahn–Banach extensions and some of its variants*, arXiv:2202.00947.

[3] S. Basu and T. S. S. R. K. Rao, *Some stability results for asymptotic norming properties of Banach spaces*, Colloq. Math. 75 (1998), 271–284.